P510/1
PHYSICS THEORY
PAPER 1
JULY 2026
3 Hours

ACEITEKA JOINT MOCK EXAMINATIONS 2026

Uganda Advanced Certificate of Education

Physics

Paper 1

Theory

Time: 3 Hours

INSTRUCTIONS TO CANDIDATES:

This examination paper consists of two sections, A and B. It has six items

Section A has two compulsory items

Section B has two parts, I and II. Respond to only one item from each part

Respond to four items in all

Any additional item(s) responded to will not be scored.

Responses to each item should start on a fresh page

Silent non-programmable scientific calculators may be used

Indicate the items responded to on the first page of the answer booklet

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SECTION A

(PARTICLES)

ITEM 1

In the factory, a worker found some sealed object left in one of the rooms some time ago. The sealed object was taken in their laboratory for investigation and it emitted entirely beta particles. Plato one of the intern and the engineer used a Geiger counter and measured an activity of 1600 counts per minute, and its half-life was 20 days from the last moment. After some time, the engineer and Plato found that the counter had reduced to 40 counts per minute, and the mass of the radioactive substance had not reduced. They wondered why and how the reading reduced and also how the count rate can be determined.

The manger found Plato asking the engineer why the emitted beta particles come from the nucleus even though the nucleus contains no electrons before decay, but she did not get a clear explanation. Near the sealed radioactive source there was a metal strip. Pato wanted to know its ability to emit radiations, so she performed a photoelectric effect experiment using a charged gold leaf electroscope which had a clean zinc plate placed on its cap. When a lamp was made to face the zinc plate, he made some observations which he failed to explain, and also wanted to know the work function of zinc metal such that he can tell other forms light which cause the same effect.

When the intensity of the light was increased, the stopping potential could not increase. Plato wondered why this was so, and when the laser pointer was directed to zinc, no observation was made, which confused Plato the more.

When Plato was moving he found students from one of the schools which had come to tour the factory talking about Rutherford's model how it's confusing to explain, and the model that makes electrons to be in specific energy levels. When Patricia heard them she asked them how the specific charge of ions that entered a magnetic field was determined and if there is any application of the effect that occurred in the scenario.

Support Material

Planck's constant $h = 6.63 \times 10^{-34}$ Js
Wavelength of light = 200 nm
Stopping potential = 2.5 V
Accelerating potential (voltage) = 400 V
Magnetic field = 0.02 T
Radius of the electron = 3 cm

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Task:

As a learner of physics at a level of senior six.

(a) Carefully analyze the scenario and help Plato to address all her concerns.

(b) Examine Patricia's concern.

ITEM 2 (MOTION AND FORCE)

In certain school, a group of students went for a tour by bus. One of the learners told a friend that the tires of the bus had flattened and he did not know why. The turn boy worked on them and they continued with the journey.

On their way to Entebbe airport, the bus moved over a hump which made them to move up and down and a learner wondered why it happened like that. When they reached the airport another learner saw an areophane and imagined how it's able to fly He could also not tell the amount of force that can lift it, however when he asked the guide about it. He said the airplane can only fly if the lift force is 6.9kN and above.

When they were coming back to school the bus slid into a trench, the driver brought a tow car to pull it. The leaners were amused on seeing the tow car pulling the bus and one asked why the tow car was not sliding. A friend said may be due to friction. One of the Students said their teacher talked about molecular explanation of friction however he could not just explain to the colleagues.

The driver said that for the tow car to be efficient, its engine should have a force of 9.7kN but he did not know how to calculate. A leaner told the driver that to calculate it, he also needs to know the coefficient of friction which confused the driver the more. Since he could not tell how to determine its value. After towing, the driver opened the bus and told the students to enter, one learner who was near him heard the door making noise and he asked how that noise can be stopped but the drive could not give an answer. When they reached school, their teacher briefed them about what they had leant and he asked Muzafaru a learner to explain any other application that uses the same principle like that of the airplane when starting to fly but he could not.

Support material

Speed of air above the wings of the areophane	$= 81ms - 1$
Speed of air below the wings of the areophane	$= 57ms^{-1}$
Total area of the wings of the aero plane	$= 3.2m^2$
Density of air	$= 1.3kgm^{-3}$
Mass of the bus	= 2000kg
Mass of the tow car	=1000kg

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The tow accelerates at rate of	$=0.8ms^{-2}$
Coefficient of friction	$= 0.25$
Acceleration due to gravity	$=9.81ms^{-2}$

Task:

As a learner of physics;

a) Analyze the scenario and address the challenges.

b) Examine Muzafaru's warry.

SECTION B

PART I (ENERGY)

ITEM 3

At a rural health centre in Kuwero, Dr. Anna and her intern Peter are struggling with a damaged solar-powered vaccine refrigerator. The refrigerator uses a thermodynamic cycle to remove heat from the vaccine compartment (maintained at 2°C) and releases it to the outside air at 27°C. The compressor does 500 J of work per cycle, and the refrigerator absorbs 1500 J of heat from the cold compartment each cycle. Peter wonders, "How much heat is released to the room" and "whether this is working as efficiently as it should" He calculates the coefficient of performance but worries because the actual performance is lower than the ideal coefficient value.

Nearby, a broken window allows humid outside air (relative humidity 80% at 27°C) to enter. The doctor explains that water vapour in the air condenses on the cold evaporator coils, forming ice. Peter could not explain why ice forms even when the air temperature is above freezing. She reminds him of latent heat of vaporization and condensation, which Peter could not determine.

To check the vaccine vials, Peter uses a compound microscope in normal adjustment but could not tell the magnification of the system. The doctor told him to use a small refracting telescope (objective 80 cm, eyepiece 10 cm focal length) to spot a distant ambulance across the valley. Peter could not tell whether the telescope produces an upright or inverted image in normal adjustment.

Later, Peter helps fix a solar panel that powers the refrigerator. The panel provides 150 W of power for 6 hours of sunlight per day, but he wanted to know whether the power it produces is enough for the refrigerator. The compressor runs 30% of the time. When Musa saw Peter, he asked how the device can be operated, which Peter failed to explain.

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Support Material

Specific Latent heat of vaporisation $L = 2.26 \times 10^6$ J/kg

Ideal gas constant for water vapour $R = 461.53$ J/kg · K

Compound microscope of object lens focal length = 1.5 cm

Compound microscope of eye piece of focal length=5.0 cm

Tube length = 18.0cm

Tasks:

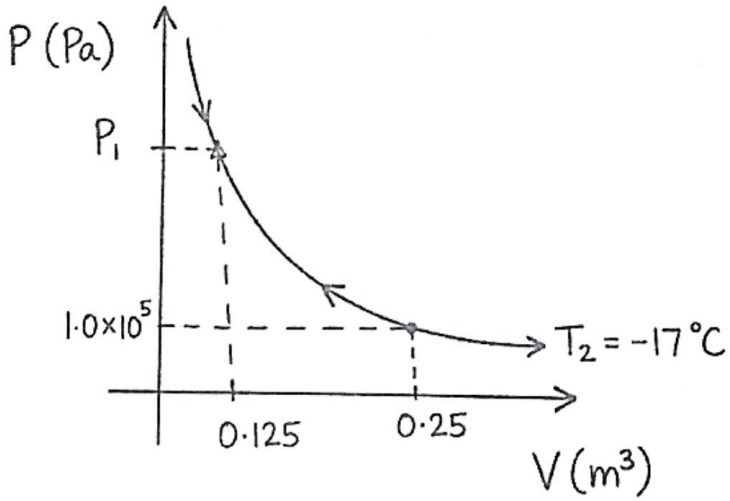
As a learner of Physics at Senior Six level:

a) Analyze the scenario to help Peter solve his puzzles.

b) Address Musa's concern.

ITEM 4

In a mountainous area of Uganda, community members face challenges leading to nose bleeding due to high air currents. A group of researchers went to this area to study the changes in atmospheric pressure and volume of the air current. The findings were plotted on a graph showing the variation of pressure with volume at a temperature of 17 °C.



The graph helped the researchers understand the variation of pressure and volume of the air current in the community. They noted that the energy carried by the air current at a given temperature of -17 °C forces community members to enter their rooms for 9 hours for safety. The members were forced to stay indoors such that the temperature

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difference between the surroundings and the inner room should be 28.1 °C for them to be safe from nose bleeding.

The researcher asked the community leader about the material of the bricks used to construct the houses and their thermal conductivity, but the leader failed to determine it.

When the researcher entered one of the members' rooms, he found them playing music using an instrument that uses a string to produce sound. He wondered how sweet the music was. Musa, the person playing the instrument, told the researcher that for the music to be nice, the string should be under tension and vibrating at a frequency of 300 Hz and above, which was the third harmonic. This confused the researcher. When Musa touched the string, it was hot, which made him wonder how heat energy is generated in the string.

Joan, one of the researchers, observed that the sound pitch produced by the string instrument depends on the variation of frequency in the string and the tension. She wanted to know how this variation can be determined.

Later, in a separate laboratory experiment, a researcher set up a sonometer to investigate the relationship between frequency and tension. A paper rider was placed on the string. The string was plucked, and the rider vibrated. When she increased the tension, the rider still vibrated but at a different frequency. She was confused because she expected a simple relationship, but the results seemed inconsistent. She also wanted to determine the speed of the wave in the string when the tension was 20 N and the linear density was 0.5 Kgm^{-1}. To understand further, she placed the sonometer in a vacuum chamber. Surprisingly, the rider still vibrated, and the frequency remained the same as in air. She could not explain why the presence or absence of air did not affect the frequency.

Support Material

- Thermal conductivity of brick wall = 0.6 $Wm^{-1}K^{-1}$
- Cross-section area of the wall = 40 cm^2
- Thickness of brick = 10 cm
- Tension in the string = 20 N
- Density of string = 600 kg/m^3
- Length of string = 1 m
- Diameter of the string = 1.03×10^{-3} m

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Task:

a) Analyze the scenario and help the researcher address their challenges.

b) Examine Joan's task.

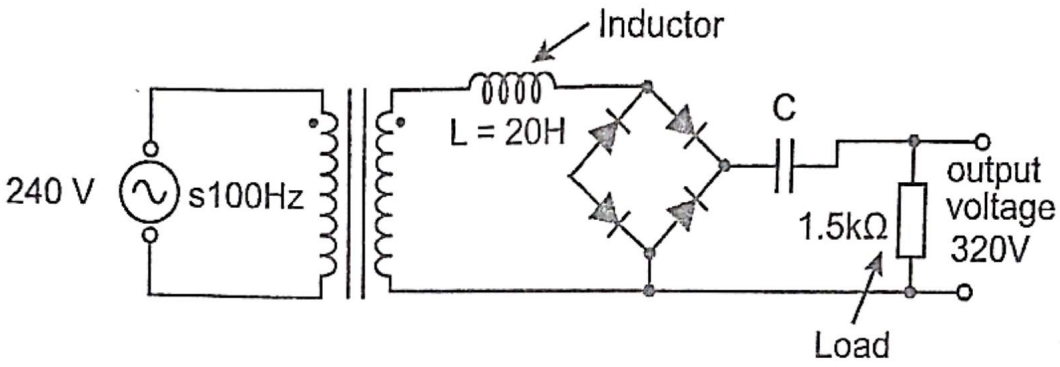
PART II

(FIELDS AND CHARGES)

ITEM 5

A certain school has a challenge of electricity fluctuations. The school management is planning to design a system which will solve the problem. To achieve this, their technician told them that they need a switch which can turn the generator on and off in case of power fluctuation, but the technician lacked the knowledge of how it works.

The circuit board of the school electricity distribution was shown in a diagram.



The technician consulted the engineer on how the circuit should be done to solve the electricity fluctuation. The engineer told him that the voltage should be improved since the hospital had many electric devices, but the technician could not explain how this can be done.

In the technician's circuit board, the technician was told to include a capacitor of capacitance 50 μF and an inductor of inductance 2H, but the capacitor was to act as a smoother, which he failed to understand.

When the technician went to the electronics shop, he wanted to buy a transformer but had no specifications, and the hospital devices required an output voltage of 400 V, but could not carry out calculations to determine it.

Marth told the technician that at the school, they had devices which could also use d.c voltage, she wondered how this could be adjusted to work on them.

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Support material

Transformer should be of output voltage of 400v and more

Task:

(a) Analyze the scenario and help the technician to address the challenges.

(b) Examine Marth' challenge.

ITEM 6

In our communities people listen to radios for many reasons among them are to promote emotional comfort, mental stimulation and to be informed of what is going on. John visited his parents with his child who was doing physics in s6. The parent complained to John of the challenge they had with radio set which often failed to tune in their favorable religious program the radio operated at a frequency of 0.755MHz John opened the radio and found it's tuning circuit consisting of a capacitor and an inductor very weak. John decided to install a new strong variable capacitor and inductor. He purchased materials and made the two small capacitor each of plate separation 2mm and maximum area of overlap $0.005m^2$ filled with a material of dielectric constant 4. The two capacitor were found in series and connected with a motor for tuning.

He made a pure inductor L from a 15cm air-cored solenoid of 100 turns and $0.012m^2$ cross sectional area John requested his child to perform the necessary calculation to establish whether the new circuit can tune in to desired channel.

John's parent also took the presence of their grandchild and asked for the explanation of the structure and mode of operation of radio charging device being used, because the grandparent had always wondered how the charger can be connected in the socket which was labeled 240 V to charge a radio whose input required only 12V

Support Material

Permittivity of free space $\epsilon_0 = 8.85 \times 10^{-12}$ F/m

Dielectric strength of air $= 3.0 \times 10^6$ V/m

Task:

As a Physics student at Senior six level:

a) Analyze the scenario and help John to address the challenges.

b) Assess the concern of the grandparent about the structure of the device.

END

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Item One

Task (a)

Decrease of activity from 1300 counts/min to 30 counts/min.

Radioactivity decay is random process where unstable nuclei emit radiations (beta particles, alpha particles and γ-rays) and transform into more stable nuclei. The activity decreases because the number of undecayed nuclei decreases over time. The mass of the radioactive substance does not change because the mass loss is extremely tiny and not detectable with ordinary balances. Additionally, the emitted beta particles are not part of the original nucleus but are created during decay.

Therefore Activity decreases due to exponential decay of unstable nuclei. Mass appears unchanged because the mass defect is too small to measure with ordinary instruments.

Time for activity to decrease

$T_{\frac{1}{2}} = 10$ days $A_0 = 1300$ $A = 30$

$$A = A_0 e^{-\lambda t}$$

$$1300 = 30\, e^{-\frac{(\ln 2)}{10} t}$$

$$t = 54.4 \text{ days}.$$

- The electrons (beta particles) are produced from energy during the weak interaction, not originally present in the nucleus. Also since the neutron in an unstable nucleus transform into a proton and an electron.

- when light is directed to the clean zinc plate placed on a gold leaf

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electroscope, the following observations are made.

When the GLE is positively charged there will be no observable change on the leaf since all the emmitted electrons are gained back to the plate.

When the GLE is negatively charged the leaf will fall show that there is loss in charge.

Work fuction

$\lambda = 200nm \quad V_s = 2.5V$

$hf = hf_0 + K.E$

$\frac{6.63\times10^{-34} \times 3.0\times10^{8}}{2.0\times10^{-7}} = W_0 + 1.6\times10^{-19} \times 2.5$

$W_0 = 3.72 eV$

Since the W_0 is prime photo electric effect will occur.

Increases in the intensity does not increase stopping potential, since stopping potential depends on the maximum K.E of the photo electron which is determined by photon energy (hf) not number of photons. Thus increase in intensity increases the number of emitted electron not kinetic energy.

- Red light photon have energy below the work function of zinc regardless of their intensity not photoelectric effect.

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Bohr's model introduces quantised energy levels. Electrons can only occupy certain orbits without radiating. They jump between orbits by absorbing or emitting photons, of specific energies explaining atomic line spectra.

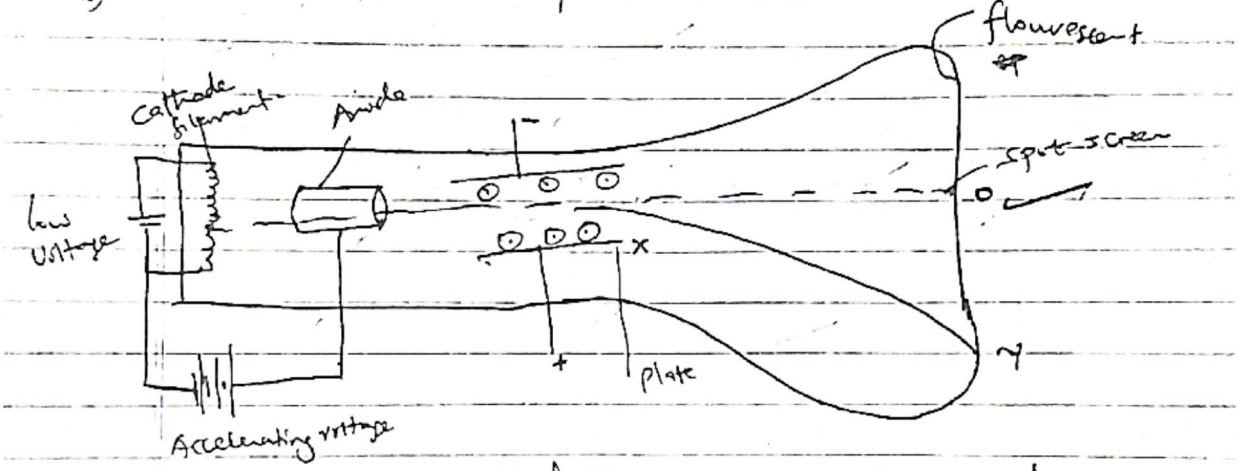
Specific charge of electron

$V = 400\,V$, $B = 0.025\,T$ $r = 3\,cm$

$$\frac{e}{m} = \frac{2V}{B^2r^2} = \frac{800}{(6.02)^2(0.03)^2}$$

$$= 2.02 \times 10^9\ C\,kg^{-1}$$

b) Specific charge determination



- Electrons are emitted thermionically from a heated filament
- The electrons are accelerated towards the fluorescent screen
- with no electric field and no magnetic fields applied at plates the electron beam strikes at the screen at point O which is noted. A magnetic field of flux density, B is applied at X to deflect the electron beam to a point Y which is noted.
- An electric field of intensity E is applied at X at right angles to the flux B at X and adjusted until the position of the beam on the screen is restored to point O.
- The pd V, the plated separation d and velocity v of the movement of the electron beam are noted

The electric force = magnetic force.

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$Bev = eE$

$v = \frac{E}{B}$ ✓

$eV_a = \frac{1}{2} m v^2$

The specific charge $\frac{e}{m} = \frac{E^2}{2V_a B^2}$ ~~$E = \frac{V}{d}$~~.

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Question 2

a) $\sigma_{steel} = \frac{F}{A_w}$

$F = \frac{\sigma_{steel}}{L} = 2.07 \times 10^{11} \times 1.5 \times 10^{-6} \times 1.6$

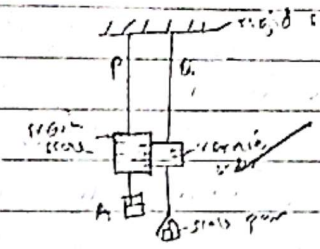
$= 4.00 \times 10^{1} N$

Copper

$F = 1.24 \times 10^{11} \times 1.5 \times 10^{-6} \times 1.6$

$= 2.11 \times 10^{1} N$

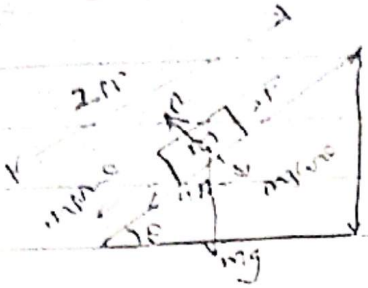
The wire is the steel wire since it can carry its load.



- Two long thin wires P and Q are suspended from the same rigid support.
- Wire P is kept taut and long wire by small mass attached at A
- The original length L of the test wire Q is measured and recorded.
- The mean diameter d is determined and the cross-sectional area $A = \frac{\pi d^2}{4}$ is found.
- Variable weights W are added to scale pan and values of the extension are noted.
- Each time the extension is noted the masses are removed from the scale pan to ensure that the elastic limit is not exceeded.
- The graph of W against e is plotted and its slope s is determined. The Young's modulus $E = \frac{6L}{4}$

- When the wire is stretched beyond elastic limit permanent displacement of atoms occurs, crystal planes slide over each other, removing the stress it does not regain its shape and size hence plastic deformation.

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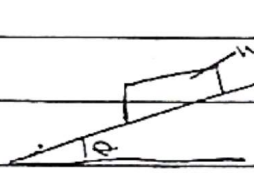
$F - \mu R - mg\sin\theta = 0$

$F = 0.25 \times Mg\cos\theta - mg\sin\theta$

$= 0.25 \times 50\times10^3\times10\cos 9 - 50\times10^3 \sin 36$

$= 39996.06N = 40{,}000N$

The wire should carry a load of 5×10^3 kg a force of 40,000 N.



weight is placed on the plane and tilted upwards until the block (weight) just starts sliding down. The angle is measured. The co-efficient of friction $\mu = \tan\theta$

$$R = \left(\frac{T^2 G m_e}{4\pi^2}\right)^{1/3} = \left(\frac{(24\times3600)^2 \times 6.67\times10^{-11} \times 6.0\times10^{24}}{4\pi^2}\right)^{1/3}$$

$= 4.24\times10^7 m$

$h = 4.24\times10^7 - 6.4\times10^6$

$= 3.6\times10^7 m$

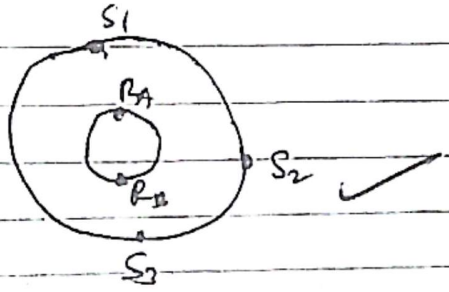
The satellite can used to transmit and receive signals

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orbital velocity

$$V = \sqrt{\frac{GM}{r}} = \sqrt{\frac{4.002 \times 10^{14}}{4.23 \times 10^{7}}} = 3076 \text{ m s}^{-1}.$$

b) A satellite can transmit signals, is addressed by the principle of electromagnetic wave transmission. The satellite receive signals from ground station via a radio frequency, processes them and retransmits them to another ground station. This allows signals travel long distances beyond the horizon



A set of three satellites are launched in a geostationary orbit. A signal from station R_A is transmitted to a satellite S_1, then S_2 and to S_3 which retransmits the to receiver at R_B.

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Item 3

Heat released to the room.

$Q_H = Q_C + W = 1500 + 500$

$= 2000\,J$

Refrigerator Coefficient of performance $= \frac{Q_c}{W} = \frac{1500}{500}$

$= 3$

Carnot co-efficient of performance $= \frac{T_c}{T_h - T_c}$

$T_c = 12 + 273 = 285$

$T_h = 27 + 273 = 300$

$= \frac{285}{32 - 285}$

Since Refrigerator Co-efficient of performance is 3 are 19.0, The refrigerator is much less efficient than ideal.

Ice forms vapour.

The evaporator coils are at temperature below 0°c. Water vapour in the humid air comes into contact with the cold coils and undergoes deposition because the dew point is above 0°c. The coil temperature is below freezing point causing water vapour to condense and freeze directly.

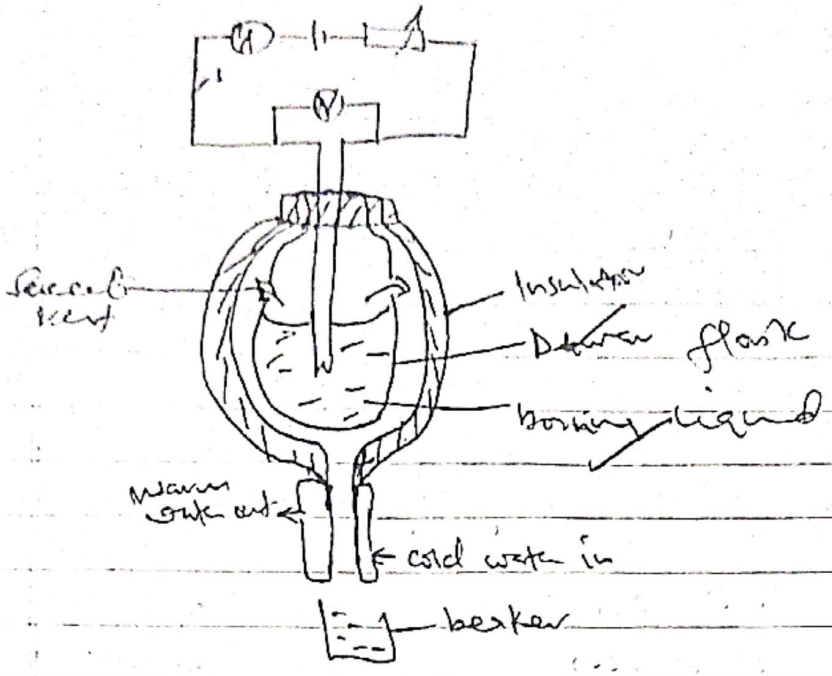
Latent heat released

$Q = mL_v$

$= 0.01 \times 2260 \times 10^3$

$= 22600\,J$

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• Switch is closed and liquid is heated until it starts boiling.

• A stop clock is started and mass m of liquid collected in time t is weighed and recorded.

• The Ammeter reading and voltmeter reading I_1 and V_1 are noted.

At steady state $I_1V_1 = \frac{m_1}{t}L_v + h$ — (1)

where h is the rate of heat loss

The rheostat is adjusted and a new Ammeter reading and Voltmeter reading I_2 and V_2 are recorded and the new mass m_2 of condensed steam collected with same t is recorded.

$$I_2V_2 = \frac{m_2}{t}L_v + h$$

The specific latent heat of vaporisation

$$L_v = \frac{I_2V_2 - I_1V_1}{\left(\frac{m_2}{t} - \frac{m_1}{t}\right)}$$

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At 27°C saturation vapour pressure = 3.56 kPa

Actual vapour pressure = 0.5 × 3.56

= 2.548 kPa

At 0°c saturation vapour pressure = 0.61 kPa

Mass condensed = $\frac{\text{Pressure difference} \times \text{Molar mass}}{RT}$

$= \frac{\quad}{8.31 \times 300}$

$= 0.018 \text{ kg/m}^3$

18g of water condenses per cubic metre of air

Compound microscope.

$f_o = 1.5$ cm $\quad f_e = 5$ cm $\quad L = 18$ cm $\quad$ Near point distance

$m = \frac{L}{f_o} \times \frac{D}{f_e} = \frac{18}{1.5} \times \frac{25}{5} = 12 \times 5 = 60$

In normal adjustment.

magnification of the microscope = 60×

The image is inverted.

- For an astronomical telescope in normal adjustment the image is inverted, for terrestrial use an erecting lens is added to make it upright.

The telescope produces an inverted image in normal adjustment.

Solar energy per day $E_s = Pxt$

$= 1.50 \times 6 \times 3600$

$= 3.24 \times 10^6$ J.

Compression run 30% of 24hrs $= 0.3 \times 24 \times 3600$

$= 25920$ J

Energy required $E_{req} = 120 \times 25920$

$= 3.11 \times 10^6$ J

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Since $3.24 \times 10^{6} > 3.11 \times 10^{6}$ the solar energy is sufficient to the panel provides enough energy.

b) Solar - powered vaccine refrigerator.

- The solar panel converts sunlight into D.C electrical energy
- A Charge controller regulates the voltage and current to prevent overcharging
- A battery stores energy for use when sunlight is absent.
- The compressor uses electrical energy to circulate refrigerant, absorbing heat from the vaccine compartment and releasing it outside A thermostat maintains the vaccine compartment $2^{\circ}C - 8^{\circ}C$. The system is automated - the compressor cycles on and off as needed powered by the battery which is continuously recharged by the solar panel during daylight hours.

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Item 04

a) Work $= P_1 V_1 \ln \frac{V_2}{V_1}$

$P_1 V_1 = P_2 V_2$

$P_1 = \frac{P_2 V_2}{V_1} = \frac{1.0 \times 10^5 \times 0.25}{0.125}$

$= 2.0 \times 10^5 \text{ Pa}$

$W = 1.0 \times 10^5 \times 0.25 \ln \frac{0.25}{0.125} = 1.73 \times 10^4 \text{ J}$

$\frac{Q}{t} = \frac{AK(\theta_2 - \theta_1)}{x}$

$\frac{1.7329 \times 10^4}{7 \times 3600} = \frac{40 \times 10^{-4} \times 0.6 \times \Delta\theta}{0.1}$

$\Delta\theta = 28.7°C$

In the room they will be safe.

Thermo conductivity of brick



- A sample disc from brick of small thickness t and diameter d is cut for brick.
- The disc is sandwiched between two metal slabs X and Y each carrying a thermometer
- steam is passed through the steam chest until the thermometers record steady temperatures θ_1 and θ_2 recorded from thermometers T_1 & T_2 respectively.

$\frac{Q}{t} = \frac{KA(\theta_2 - \theta_1)}{t}$

- The sample is withdrawn and slab Y is heated directly when in contact with X until its temperature

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rises above θ_2 by 10°c
- The steam chest is removed and the disc is placed on top of slab y
- The temperature of the slab y is recorded at suitable time intervals.
- A cooling curve is plotted and the slope s of the graph at θ_1 is determined.
- The mass m of slab y of specific heat capacity c is determined $\frac{Q}{t} = mcs$

The thermal conductivity K of the disc can be determined from $mcs = \frac{K\pi d^2(\theta_2 - \theta_1)}{4t}$

From this the thermal conductivity of brick can be known and obtained which the researcher gets its value.

Music from string instrument

fundamental note or frequency f_0

$f_0 = \frac{1}{2l}\sqrt{\frac{T}{m}}$ $\quad m = \rho x$

$= \frac{1}{2(1)}\sqrt{\frac{20 \times 1}{0.000499994}}$ $\quad = \frac{600 \times \pi (1.03 \times 10^{-3})^2 \times 1}{4}$

$= 100.0 Hz$ $\quad = 0.000499994 kg$

for Third harmonic $f = 3f_0 = 3 \times 100$

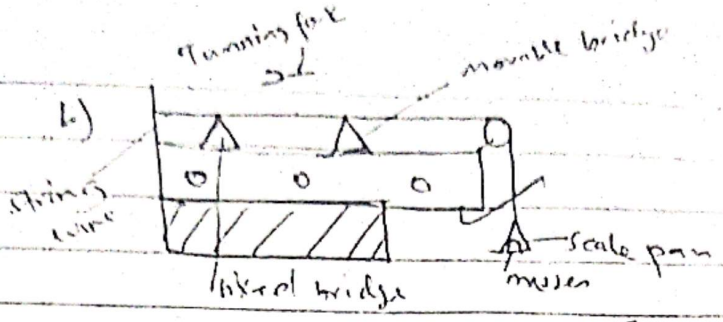
$= 300 Hz$

The string can produce more good note and nice music.

The higher the harmonics, the more produced becomes nice

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b)

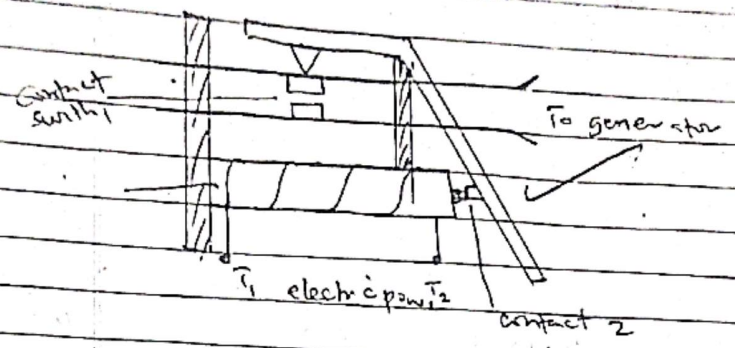


- The length between the two bridges is kept constant.
- A suitable mass m is attached to the free end by a string.
- Pluck the string in the middle and a tunning fork of known frequency f is sounded near it
- Vary the mass m on the scale pan until when a loud sound is heard.
- Record the mass m and its corresponding frequency f in a suitable table including values of f^2 and $T = mg$
- Repeat the procedures using different tunning forks of different frequency.
- Plot a graph of f^2 against T. It's straight line through the origin implying that $f^2 \propto T$. Thus frequency increases with increase in Tension.

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Item 5

a) The switch which can do the relay Switch.



when electric power is on the current flows through the core and it gets magnetised to attract the armature and the contact switch, happens then the generate turns off.

when current is off the contact 2 open since the armature off the core to path the content switch, 1 and the generate works.

Input voltage and output voltage

$V = IR$

$320 = I\,(1.5\times10^3)$

$I = \frac{320}{1.5\times10^3} = 0.2133\,A$

Inductance

$X_L = 2\pi fL = 2\pi \times 100 \times 20$

$= 12566.37\,\Omega$

Capacitance $X_c = \frac{1}{2\pi fC} = \frac{1}{2\pi \times 100 \times 50 \times 10^{-6}}$

$= 31.831\,\Omega$

$Z = \sqrt{X_L^2 + X_c^2}$

$= \sqrt{12566.37^2 + 31.831^2}$

$= 12566.4\,\Omega$

Effective resistance $= 12566.4 + 1500$

$= 14.0664\,k\Omega$

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$V = IR$

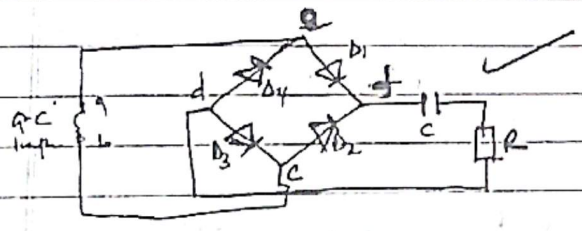
$= 0.2 \times 12.2 \times 14066.4$

$= 3000\ V$

The transformer used should output voltage 3000 V to improve the efficiency of the hospital.

- The capacitor that acts as smoother to allow current flow through it all the time since it stores charges in such that there is not power fluctuations at the hospital.

b) Martha's challenge



when a is made positive with respect to b diode D_1 and D_3 will conduct since D_2 and D_4 have high resistance and current flow in the direction a e f R d c b through the load R. When b is made positive with respect to a diodes D_2 and D_4 conducts and current flow in the direction b c f R d e a through the load R in the same direction thus ac has been converted to d.c and the horizontal device that was d.c can work.

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Item 6

$f = 5.2 \times 10^6$ Hz

$A = 0.005\ m^2$ $\quad d = 2\ mm = 0.002\ m$

$K = 4$ $\quad \varepsilon_0 = 8.85 \times 10^{-12}\ F/m$

$$C = \frac{K\varepsilon_0 A}{d} = \frac{4 \times 8.85 \times 10^{-12} \times 0.005}{0.002}$$

$$= 8.85 \times 10^{-11}\ F.$$

Since $C_1 = C_2 = C$

$$C = \frac{C_1 C_2}{C_1 + C_2} = \frac{C}{2} = \frac{8.85 \times 10^{-11}}{2}$$

$$= 4.425 \times 10^{-11}\ F$$

Inductance of solenoid

$l = 15\ cm = 0.15\ m$ $\quad N = 100$ $\quad A_T = 0.012\ m^2$

$\mu_0 = 4\pi \times 10^{-7}$

$$L = \frac{\mu_0 N^2 A_T}{l} = \frac{4\pi \times 10^{-7} \times 100^2 \times 0.012}{0.15}$$

$$= 1.005 \times 10^{-3}\ H$$

$$f = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{(1.005 \times 10^{-3})(4.425 \times 10^{-11})}}$$

$$= 7.55 \times 10^5\ Hz$$

$$= 0.755\ MHz$$

Since the $f = 0.755$ MHz is not 0.52 MHz the circuit will not tune to the desired channel with fixed capacitors.

A variable capacitor is needed to adjust to exactly 0.52 MHz.

required capacitance for 0.52 MHz

$$f = \frac{1}{2\pi\sqrt{LC}} \Rightarrow C = \frac{1}{4\pi^2 L f^2}$$

$$= \frac{1}{4\pi^2 (1.005 \times 10^{-3}) \times (5.2 \times 10^5)^2}$$

$$= 9.32 \times 10^{-11}\ F$$

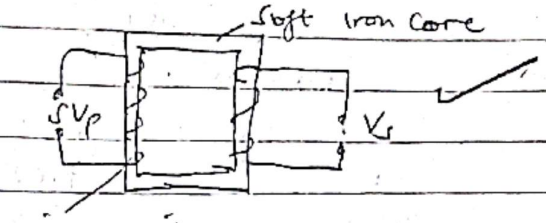
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So John needs a Variable capacitor that can reach 93pF.

b)

Step-down transformer to reduce 240V to 12



Alternating voltage connected to the primary coil with more number turns produces an alternating current in it. This setup an alternating magnetic flux in the core which links up the secondary coil and thus induces an e.m.f in the coil. The magnitude of the emf induced in the secondary coil is $V_s = N_s A \frac{d\phi}{dt}$

The changing magnetic flux also links the primary and induces a back emf in the same coil where $V_p = N_p A \frac{d\phi}{dt}$. The transformer should be with N_p/N_s giving $V_p > V_s$.

→ Rectifier explanation or
→ AC/DC Adapter explanation of how it works.

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